Reinforcement Learning-based Analog Circuit Optimizer using g_m/I_D for Sizing

Minjeong Choi
Samsung Advanced Institute of Technology
Suwon, Korea
minjj00.choi@samsung.com

Youngchang Choi
Electrical Engineering
POSTECH
Pohang, Korea
yc.choi@postech.ac.kr

Kyongsu Lee
Electrical Engineering
POSTECH
Pohang, Korea
kyongsul@postech.ac.kr

Seokhyeong Kang
Electrical Engineering
POSTECH
Pohang, Korea
shkang@postech.ac.kr

2023 60th ACM/IEEE Design Automation Conference (DAC) | 979-8-3503-2348-1/23/$31.00 ©2023 IEEE | DOI: 10.1109/DAC56929.2023.10247739

***Abstract*—Designing analog circuits incurs high time costs because designers must consider numerous design variables or trade-off relationships of circuit performance based on a lot of knowledge and experience. To reduce design time, various machine learning methods have been used to optimize analog circuits by learning the correlation between the device size and the circuit performance. However, it is difficult to train the correlation because of its high non-linearity and wide design space. In this paper, this study proposes a new framework to optimize analog circuit designs by combining reinforcement learning (RL) and the sensitivity analysis with g_m/I_D sizing, which is more intuitive for interpreting circuit performance. Furthermore, the universal value function approximator (UVFA), previously proposed in RL, is modified more simply to make it easier to find the target design. Additionally, the dataset is rearranged and sampled by the criteria that are established based on the principle of circuit operation, which helps to orient the agent to learn the circuit operation. Using the proposed methods, we optimize three types of differential amplifiers with common mode feedback circuits and obtain the best circuit design. Compared to baseline, we find the optimal point using modified UVFA, and moreover, reduce the number of iterations by 42.2%, 39.5%, and 37.5%, respectively, for the three test cases.**

***Index Terms*—Analog circuit sizing, Automated design, Efficient search, Reinforcement learning**

I. Introduction

As CMOS technology continues scaling, electrical properties such as power consumption and operating speed are improving. However, designing analog integrated circuits has become more challenging because the design rules become more complex [1] and also, in the short channel, the drain current equation induced in the long channel is inconsistent with actual device operation [2]. For these reasons, custom-designing of analog circuits by sizing is required significant design experience and knowledge of device characteristics by performing a large number of simulations. Therefore, design methods have been researched to reduce trial and error. In response, there are several studies that use machine learning instead of adjusting circuits manually. Regarding this approach, bayesian optimization (BO) [3], [4] and reinforcement learning (RL) [5]–[8] were used to optimize analog circuits with sizing.

The BO method computes the surrogate model of the objective function using a Gaussian process (GP) based on probability, which has the advantage of avoiding unnecessary repetitive searches. However, the computational cost is high when calculating the mean and covariance at each data point using the GP regression. The training and prediction complexity is O(N^2) and O(N^3), respectively, where N is the number of data points [9]. Therefore, as the amount of data accumulates, BO requires a longer time to obtain the same number of samples. Compared to BO, RL obtains data samples almost constantly, regardless of data accumulation [6], thereby reducing the runtime. RL studies [5]–[8] train the agent to ensure that the circuit performance reach the target specification based on the geometric design variables of circuits. However, geometric design variables have a high non-linearity pertaining to the performance of the circuit, therefore it is not easy to find optimal points in a wide design space from the perspective of the learning agent.

In this paper, we propose a novel framework for exploring targeted analog circuit design with RL integrating the g_m/I_D sizing [10]–[12], which is closely related to the performance of the circuit. The g_m/I_D sizing method has the advantage that analog circuit designers can design without considering the short channel effect [2] instead of using the drain current equation to design. This is because the current efficiency g_m/I_D, one of the transistor characteristics, is not a theory-induced equation but is obtained from experiments. Furthermore, we performed a sensitivity analysis between the g_m/I_D of the building block of the circuit and the performance of the circuit to devise the search method of the optimal point using the g_m/I_D with deep deterministic policy gradient (DDPG) algorithm [13] as a baseline. In addition, we applied universal value function approximators (UVFA) [14] to optimize the circuit design by combining the state and a flag vector, which represents the circuit performance. Finally, we performed sampling to train the network while considering the prior knowledge of the circuit design to increase the data efficiency. The data set is listed according to certain criteria and takes one part of the data samples, each from the bottom and top for training, instead of random samples. The contributions of our study can be summarized as follows:

- We propose a RL-based analog circuit optimizer with fast convergence, which uses a sensitivity analysis between g_m/I_D values and circuit performance.
- We introduce simplified UVFA method which has smaller state space and make it easier to find the target design, while preserving the good performance.
- In our RL-based analog circuit optimizer, we present new sampling methods based on the *knowledge of circuit design*, which enables the RL agent explore the data efficiently.

With these three contributions, we reduced the number of training iterations and runtime compared with the DDPG algorithm to find the optimal point that satisfies the target specification. The remainder of this paper is organized as follows. Section II explains the background of RL and the method of g_m/I_D sizing and describes related works. Section III presents the proposed work and algorithm and Section IV shows the experimental results for the three circuits. Finally, Section V concludes and summarizes the study.

II. Preliminaries

Sections II-A through II-C explain the basic background of RL, and Section II-D describes the analog circuit design methodology. Section II-E interprets previous studies related to RL and g_m/I_D sizing.

A. Reinforcement learning

RL is the process by which an agent learns in an unknown environment, such as a game. When an agent takes action A in state S, it receives a reward R by interacting with the environment. In more detail, at the point of time t, if action a_t is selected according to the policy $\pi : S \rightarrow A$ in the current state s_t, the action is taken and reaches the next state s_{t+1}. At the same time, the reward is obtained through the reward function, $r : S \times A \rightarrow R$. The agent is trained in the direction of maximizing the expected value of $R_t = \sum_{i=t}^{\infty} \gamma^{i-t} r(s_t, a_t)$, which is the discounted sum of the future reward, where the discounting factor is $\gamma \in [0, 1]$. The action-value function is $Q^{\pi}(s_t, a_t) = \mathbb{E}[R_t|s_t, a_t]$, and the goal is to find the maximum value of the action-value function.

RL can primarily be divided into on-policy and off-policy algorithms. The on-policy algorithm directly evaluates the policy to select an action and is straightforward to implement. In contrast, the off-policy algorithm learns the target policy, which is to be finally learned apart from the current policy that selects the action. Furthermore, it can re-evaluate previous experiences.

B. DDPG algorithm

The DDPG algorithm [13] is a representative method for off-policy reinforcement learning. While learning the unknown environment, it stores tuples (s_t, a_t, r_t, s_{t+1}) in a replay buffer at every step, and uses an experience replay that samples prior experiences in a buffer as much as the batch size. It consists of an actor network that can select continuous actions and a critic network that evaluates the value of the action.

C. Universal value function approximator (UVFA)

The value function evaluates the expected reward from the current state. UVFA [14] is a method for evaluating value function by combining the state and goal. If G is defined as a possible set of goals, i.e., g, these goals are defined as the inclusion relation of the state ($g \subseteq S$). To rephrase, goals act as a flag for the state. When the concept of a goal is added to DDPG, the definition can be rewritten as follows: The policy is $\pi : S\times G \to A$, the reward is $r : S\times A\times G \to R$ and action-value function is $Q^{\pi}(s_t, a_t) = \mathbb{E}[R_t|s_t, a_t, g]$. It has been proven that high-performance approximators can distinguish these states and goals well.

D. g_m/I_D sizing methodology

In short channels, the drain current formula derived from long channel is no longer followed due to short channel effects, leading to design difficulties. To overcome difficulties, studies have attempted to design with the g_m/I_D sizing method [10]–[12]. The g_m/I_D, the ratio of transconductance to the DC drain current, is one of the characteristics of a transistor measured in a process. This value is a measure of the efficiency of the conversion of the current I_D into the transconductance g_m, which represent power and gain, respectively.

In g_m/I_D sizing method, the designer sets g_m/I_D from empirical data (Fig. 1) and approximately determines the device size from the g_m/I_D value. In particular, the g_m/I_D value is large when operating with low power in the weak inversion area and small when operating at high speed in the strong inversion area. Therefore, after setting the operating area according to the target, the size can be determined using the lookup table (LUT) obtained from the measurement graph of g_m/I_D versus $I_D/(W/L)$. (Fig. 1) The $I_D/(W/L)$ is the normalized drain current, which enables designers to determine the width once the length and I_D values are given. The LUT can be measured experimentally with a single NMOS or PMOS.

The g_m/I_D value is intuitive because it is directly related to the performance of the circuit, as well as being able to determine the operating region. For example, the intrinsic gain stage comprising a single saturated transistor, the DC gain A_{DC} is induced as in Equation (1), where V_A is the early voltage. It can be simply seen that the intrinsic gain has a linear relationship with g_m/I_D.

$$A_{DC} = -\frac{g_m}{I_D} V_A \tag{1}$$

E. Previous works

When automating analog circuit design based on reinforcement learning, DDPG [13] was primarily used as a baseline for a continuous design space search. However, due to the vast design space, searching the optimal point of an analog circuit design using only the DDPG algorithm is difficult. Therefore, it has been studied to propose new techniques rather than using only the baseline itself [5], [6], [8]. Most studies set state S to the size of the design variables, and each of them

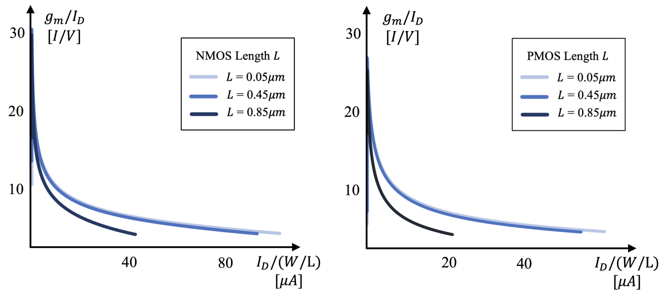


Fig. 1. g_m/I_D value obtained by sweeping the current at a fixed width and length of MOSFET. The g_m/I_D value is plotted according to the change in length L at a fixed width W. (a) Characteristics of NMOS (b) Characteristics of PMOS

takes an increasing or decreasing action A, so that they are learned to have a higher expected reward value.

In other studies, Artificial Neural Network (ANN) models were trained to obtain design variable solutions from circuit performance. Among these, [15] study that trained ANN model by generating data based on g_m/I_D showed higher predictive accuracy compared to device size-based data. These experimental data demonstrate that when neural networks learn circuits, g_m/I_D is much better to learn than the device size. Furthermore, the ANN model obtained by g_m/I_D-based data is applied to the RL [16]. However, this framework comprises a sparse reward environment: although it has been developed into a hindsight experience replay [17], it is not necessary to force a sparse reward environment. Therefore, we train the RL agent with g_m/I_D-based data, but propose an efficient optimization method framework rather than a sparse reward environment.

III. Proposed method

Section III-A and III-B explain basic definitions and setup for RL, and Section III-C represents the values to be calculated before training the RL agent. From Section III-D to Section III-F, the proposed methods are shown with Fig. 2.

A. Problem definition

When designing analog circuits, after the topology is determined, simulations are performed to satisfy the target specification with sizing. The problem we attempt to solve can be defined as:

$$\boldsymbol{\chi} = \underset{x \in \mathbb{X}^{N\times M}}{\operatorname{argmax}} \; FoM(x) \tag{2}$$

The *FoM*, Figure of Merits, is an objective function to maximize, detailed in equation (8), expressed by all parameters to be optimized, such as gain, phase margin, power consumption, and so on. The $\mathbb{X}^N$ represents design variables, and $\mathbb{X}^{N\times M}$ is the design space. From these variables, circuit performance is obtained and FoM is calculated. The targeted solution for sizing, which is $\boldsymbol{\chi}$, can be derived from the maximum *FoM* value. In this study, we propose an efficient search method to obtain sizing data that satisfies the target specification.

B. Reinforcement learning setup

The DDPG algorithm is used as the baseline to define a continuous action and re-evaluate prior learning, which is an advantage of the off-policy algorithm. Based on this baseline, it was intended to reflect how analog circuit designers adjust the size with g_m/I_D sizing method. In the t episode, in a given circuit, the state s_t is specified in terms of the g_m/I_D of the building block, the bias current I_B, the length of transistor L, resister Res and capacitor C, and the action is taken as much as the amount of each value increases or decreases. The number of each of these components, i, j, k, l, and m, is defined as much as they are in the circuit.

$$s_t = [\cdots,\ g_m/I_{D_i},\ I_{Bj},\ L_k,\ Res_l,\ C_m,\ \cdots] \tag{3}$$

$$a_t = [\cdots,\ \Delta g_m/I_{D_i},\ \Delta I_{Bj},\ \Delta L_k,\ \Delta Res_l,\ \Delta C_m,\ \cdots] \tag{4}$$

The reward r_t is set as the difference between the predicted next state and the current state at the current episode, and the current state represents $FoM(x_t)$, as defined in Equation (8).

$$r_t = FoM(x_{t+1}) - FoM(x_t) \tag{5}$$

In [7], if the circuit performance to be optimized satisfies the given target specification, the FoM function value converges to zero. In our research, we modified the FoM expression using parallel translation of that function to allow each parameter to converge to zero faster as it approaches each target. The modified FoM expression is specified in (6), (7), and (8) using T_{spec}, which represents the target specification.

$$F_{plus}(p_+) = \begin{cases} \sum\limits_{p_+ \in P_+} \alpha^{\frac{-p_+ + T_{spec}}{p_+ - T_{spec} - 1}} & if\ p \leq T_{spec} \\ 0 & if\ p \geq T_{spec} \end{cases} \tag{6}$$

$$F_{minus}(p_-) = \begin{cases} \sum\limits_{p_- \in P_-} \alpha^{\frac{-p_- + T_{spec}}{p_- - T_{spec} - 1}} & if\ p \geq T_{spec} \\ 0 & if\ p \leq T_{spec} \end{cases} \tag{7}$$

$$FoM(x) = \sum_P (F_{plus}(p_+) + F_{minus}(p_-)) \tag{8}$$

The circuit performance set P that we target consists of P_+ set and P_- set. The P_+ is a set of parameters that indicate better performance as the value increases, such as gain and CMRR. The P_- is a set of parameters that indicate better performance as the value decreases, such as the power consumption and the settling time. Parameters belonging to set P_+ follow $F_{plus}(p_+)$ function (6), and parameters belonging to set P_- follow $F_{minus}(p_-)$ function (7). The α determines the rate at which it converges to zero, which can be considered as a weight for its parameter and fixed to 1. Consequently, the FoM expression (8) is defined by combining all the function values corresponding to each target parameter.

C. Pre-calculation of g_m/I_D LUT and sensitivity analysis matrix

Data is extracted according to length L from the g_m/I_D graph for $I_D/(W/L)$ (Fig. 1), and the data is stored in the form of LUT, both NMOS and PMOS. Since the variation in the g_m/I_D value is small depending on the width, the smallest and largest values within the width range were simulated, and the average g_m/I_D value was stored in the LUT. This LUT can be used to calculate the width if the current and the channel length are fixed.

Then, a sensitivity analysis was performed to determine the extent of change in circuit performance according to the variable defined in the state. The variance-based sensitivity analysis was conducted using the Sobol method, which can measure in a nonlinear environment. As a result, it can be expressed in a matrix how sensitive the various circuit performances are for each design variable, included in the state defined in (3).

D. Actor network reflecting sensitive analysis

If the sensitivity analysis matrix obtained in advance is designated as A, the performance of the circuit is $p_i \in P$, and the target of its circuit performance is $T_{p_i} \in T_{spec}$, the following proportional expression is satisfied, where $\mathbb{X}^N$ is design variables defined in the state.

$$\frac{T_{p_i} - p_i}{p_i} \propto A \cdot \frac{\mathbb{X}^N_{predicted} - \mathbb{X}^N}{\mathbb{X}^N} \tag{9}$$

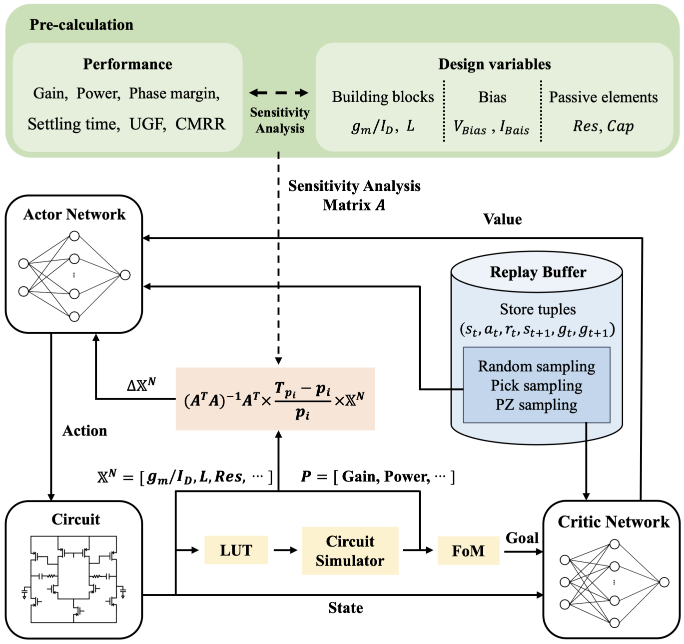


Fig. 2. The overall framework of proposed method

$$\Delta\mathbb{X}^N := (A^TA)^{-1}A^T \cdot \frac{T_{p_i} - p_i}{p_i} \cdot \mathbb{X}^N \propto \mathbb{X}^N_{predicted} - \mathbb{X}^N \tag{10}$$

Through this proportional expression, we can see the interrelationship of how much the design variable should be changed, which is represented $\mathbb{X}^N_{predicted} - \mathbb{X}^N$, to obtain the circuit performance target. The $\mathbb{X}^N_{predicted}$ means the predicted state to reach the target specification. Instead of putting the state itself in the input of the actor network, the network was trained by putting the amount of change calculated by sensitivity analysis as input. That is, based on the sensitivity analysis matrix, the next state is roughly predicted, and the actor network is trained to reduce errors so that it can be predicted more accurately within the given range.

E. UVFA for circuit optimization

We applied the UVFA method [14], which uses the concatenation of the goal and state as the input value for the value function. Since a new set of goals had to be defined, we proposed to define a set of goals as the FoM function obtained from the state to apply UVFA to our work. As explained in Section II-C, a goal acts as a flag from a state perspective and can be defined as a subset of states. However, to apply this to a process that optimizes analog circuits, the number of state elements must be large, making it difficult and complex to define a goal. Therefore, we define the goal as the FoM value, which can be represented by one vector corresponding to the state and can also act as a flag. In other words, the high-dimensional design space of the state is mapped to one dimension of the FoM, making UVFA simpler to apply.

Eventually, the RL agent learns the convergence of the FoM function to zero. The FoM value obtained from the state in each episode can be a subset of the goal G, which represents the state as a vector. Therefore, in each episode, we store tuples $(s_t, a_t, r_t, s_{t+1}, g_t, g_{t+1})$ in the buffer so that the RL agent can observe the trajectory of how the goal changes depending on the state, leading to more effective learning than in the existing baseline.

F. Novel sampling method

The RL agent is trained with an actor network and a critic network, using batch sizes of randomly sampled data. In the sampling process,

Algorithm 1: Search Optimal Design Method

```
Requires: FOM values from state F(x), Batch size N,
          Random sampling section S
Initialize replay buffer R
Set s_0 with random samples and get g_0 from s_0
F_max = F(s_0)
for episode = 1, T do
    Select action a_t ← π(s_t||g_t)
    Update new state s_{t+1} ← s_t + a_t
    Get reward r_t ← F(s_{t+1}) - F(s_t)
    F_max = max(F_max, F(s_{t+1}))
    Store transitions (s_t, a_t, r_t, s_{t+1}, g_t, g_{t+1}) in R
    if episode < S then
        Sample N transitions (s_i, a_i, r_i, s_{i+1}, g_i, g_{i+1}) from R
          randomly
    end
    else
        Sample N transitions (s_i, a_i, r_i, s_{i+1}, g_i, g_{i+1}) from R
          with Random or Pick or PZ sampling
    end
    Perform an optimization using DDPG algorithm
end
```

we propose an efficient method to select data according to specific criteria based on prior knowledge of circuit design. We sort the data in the buffer according to two criteria and then randomly sample by half the batch size at the top and bottom, respectively, within a certain range. This is because the state and goal are concatenated as the input of the network: the agent learns a clear contrasting device characteristics that strongly influence the goal. The criteria used in this study are as follows.

The first classification criterion is transistor channel length, and will be referred to as *"Pick sampling."* By analyzing the gain of the amplifier circuit, the transistors connected to the output port have a large influence, which is calculated in the form of $g_m r_0$. This expression is approximately proportional to the length and inversely proportional to the overdrive voltage of the transistors. Therefore, if classified as the length of the transistor, the gain can be well distinguished. Furthermore, because gain has a trade-off relationship with speed or power, it can also be a well-distinguished measure.

The second classification criterion is the aspect ratio W/L of the transistor, as seen from the output, which is designated based on the location of the poles and zeros. This classification will be referred to as *"PZ sampling."* The positions of the poles and zeros are crucial for determining gain and phase graphs as a function of frequency. Although the correct pole and zero should be theoretically calculated from the transistor's $r0$ calculation, it is classified based on an approximate aspect ratio, i.e., width/length, to make it easier for RL agent to learn. Additionally, we chose this criterion because g_m/I_D makes an LUT for $I_D/(W/L)$.

These criteria are based on an approximately proportional relationship, however this information can help with training when performing sampling. In each episode, we trained the network by mixing random, Pick, and PZ sampling methods and selected data with batch sizes.

IV. Experimental setup and Results

A. Experimental setup

The *Samsung-28nm* technology was used and we optimized three amplifier circuits: a basic two-stage amplifier, a telescopic amplifier [18], [19], and a folded cascode amplifier [20]. These three circuits are configured as fully differential amplifiers connecting an additional common mode feedback (CMFB) circuit. We fixed $VDD = 1V$ and set the size range of the circuit elements as follows: the width of the transistor is from $0.08\mu m$ to $5\mu m$, the length of the transistor is from $0.03\mu m$ to $1\mu m$, the capacitor is from $0.1pF$ to $50pF$, the load capacitor is from $1fF$ to $1pF$, the resistor is from $1k\Omega$ to $1M\Omega$, and the bias voltage is from $0V$ to $1V$.

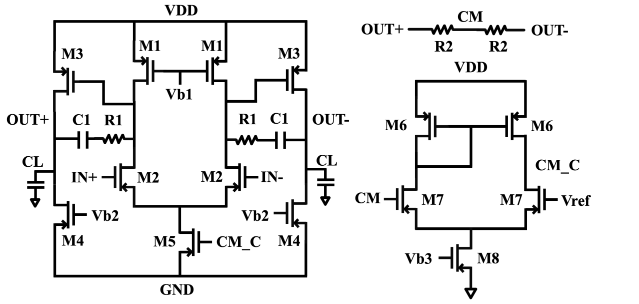


Fig. 3. Basic two-stage amplifier with CMFB circuit

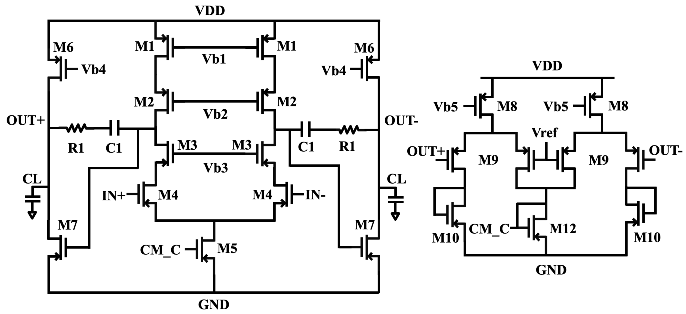


Fig. 4. Telescopic amplifier with CMFB circuit

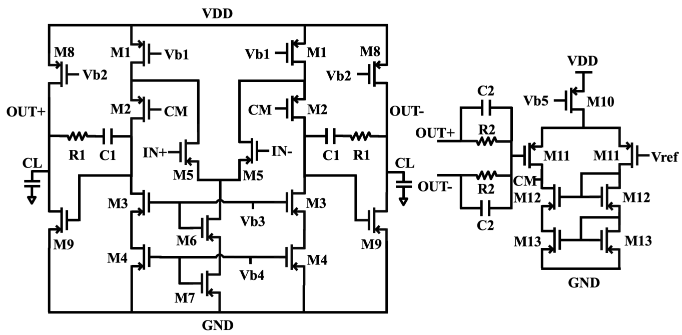
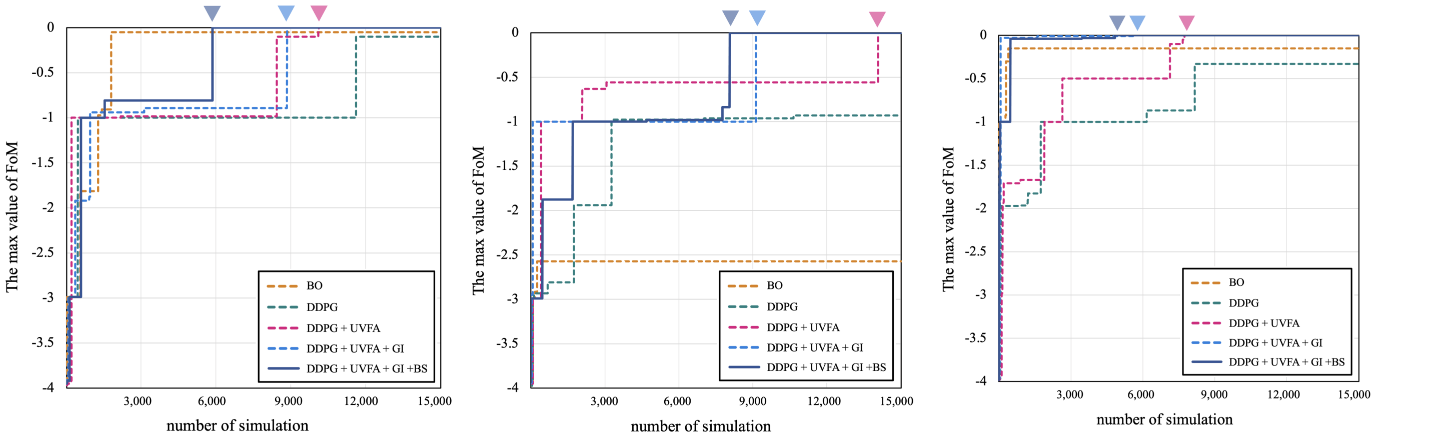


Fig. 5. Folded cascode amplifier with CMFB circuit

Before training the agent, the LUT of g_m/I_D was constructed from a saturated single transistor within that range. Plus, the sensitivity analysis of the circuit performance was performed on the g_m/I_D and length of the building blocks, passive elements, bias current, and bias voltage, which were to be designated as the state for each circuit. These values for the state were normalized from -100 to 100 and the action was selected from -1 to 1. We especially optimized 6 parameters of circuit performance: gain, phase margin (PM), power consumption, unit gain frequency (UGF), settling time, and common mode rejection ratio (CMRR). For each circuit, a target was set for each performance, and the criteria for extracting from the buffer were set through a simple calculation equation.

First, the basic two-stage amplifier has 23 elements to be optimized, including the CMFB circuit (Fig. 3) that contains building blocks of g_m/I_D and L from M1, M2, M3, M4, M6, and M7, I_D and L from M5 and M6, and other elements such as capacitor, resistor, and bias voltage. The target parameters are $20dB$ for gain, 50° for PM, and $30\mu W$ for power consumption, $10MHz$ for UGF, $1\mu s$ for setting time, and $60dB$ for CMRR. The purpose of this circuit was to ensure stability and have an appropriate gain with low power. The two-stage configuration of the circuit is to increase the gain, and the differential



(a) Basic two-stage amplifier (b) Telescopic amplifier (c) Folded cascode amplifier

Fig. 6. The result of the maximum value of FoM according to the number of simulations. The inverted triangle mark indicates convergence points.

output is to increase the voltage swing range. The gain can obtain $g_{m1}(r01//r02)$ in the first stage and $g_{m4}(r03//r04)$ in the second stage, resulting in a total gain of $g_{m1}(r01//r02)g_{m4}(r03//r04)$. After all, transistors that affects gain and location of poles and zeros very directly is theoretically M1, M2, M3, and M4, connected to the output of stage1 and stage2. Therefore, criteria of *Pick sampling* and *PZ sampling* is composed of length and width/length of transistor, from M1 to M4. In addition, the amplifier gain of the CMFB circuit should be sufficiently high, so M6 and M7 is also included in the criteria for sampling in the buffer in the same way.

Second, the telescopic amplifier has 30 components to be optimized (Fig. 4), which are considered building blocks and current bias, like the basic circuit. The target parameters are $55dB$ for gain, 50° for PM, and $0.5mW$ for power consumption, $1MHz$ for UGF, $0.1\mu s$ for setting time, and $60dB$ for CMRR. This circuit has a gain calculated as the form of $g_m r0^2$, and is used to boost gain by stacking transistors. Therefore, the gain was targeted relatively high. As described in the previous basic circuit, the resistance viewed from the output of each stage is important. Based on this principle, we select the length and width/length of M1, M2, M3, and M4 seen at the output of the stage1 and the M6 and M7 seen at the output at the stage2 for sampling.

Third, the folded cascode amplifier has 34 components to be optimized (Fig. 5), and the target parameters are $50dB$ for gain, 50° for PM, and $50\mu W$ for power consumption, $3MHz$ for UGF, $1\mu s$ for setting time, and $60dB$ for CMRR. In this circuit, gain is boosted, but since $r0$ of input is also visible on the output side, which reduces gain by connecting this resistor in parallel at output. Therefore, the target gain decreased relatively, but the power was also set low. With the same principle as the telescopic amplifier, the resistance of M5 also can be seen from the output side of the stage1. As a result, from M1 to M5 at the stage1 and M6 and M7 at the stage2 are selected in a buffer for criteria that its length and width/length.

B. Simulation setup

First of all, experiments were conducted to compare BO and RL, and the BO was limited to up to 10 hours for fair comparison with RL because it takes longer as data accumulates. On average, the RL method we proposed takes about 1.8 seconds per episode. In addition, the RL network was trained based on random sampling, so we conducted the experiment several times, and among them, attached an average graph that shows well the tendency of the algorithm when observing it. In each episode, W and L were calculated using LUT to obtain the performance of the circuit from states such as g_m/I_D and I_D set in Section IV-A for each circuit. Then, when *Hspice* simulation

TABLE I
TARGET AND FINAL RESULT VALUES FROM SIMULATIONS

Performances	Basic two-stage amplifier		Telescopic amplifier		Folded cascode amplifier	
	Target	Result	Target	Result	Target	Result
Gain (dB)	20	33.4	55	57.3	50	69.5
PM ($^\circ$)	50	71.73	50	87.8	50	81.5
Power (mW)	0.003	0.0028	0.5	0.038	0.05	0.035
UGF (MHz)	10	75.1	1	15.5	3	4.73
Settling time (μs)	1	0.001	0.1	0.005	1	0.97
CMRR (dB)	60	88.1	60	105	60	111

was performed and goal is obtained through FoM, the reward was calculated and networks were trained. The batch size was fixed to 128.

Regarding to RL, we observed simulation results by applying the proposed methods in Section III with the target specifications specified on Table I for each circuit. We applied simplified UVFA described in Section III-E, g_m/I_D method (GI) in Section III-C and III-D, and buffer sampling (BS) in Section III-F. To show the effectiveness of each methods, we conducted five experiments: BO, DDPG, DDPG+UVFA, DDPG+UVFA+GI, and DDPG+UVFA+GI+BS. Note that BO, DDPG and DDPG+UVFA were optimized based on geometric design parameters, such as width and length of transistor, according to the prior works, while DDPG+UVFA+GI and DDPG+UVFA+GI+BS were optimized based on electrical design parameters.

In the case of the BS experiment, up to the initial 1,500 episodes (S = 1,500 on Algorithm 1), the RL agent is trained with randomly sampled data. After that, random sampling, Pick sampling, and PZ sampling are alternately applied to certain section of the episode. The criteria of *Pick sampling* and *PZ sampling*, length and aspect ratio, have been set based on Section IV-A.

C. Result and analysis

We conducted five experiments on each circuit, and finally, the detailed circuit performance values obtained by our proposed DDPG+UVFA+GI+BS method are shown in Table I. Table II and Fig. 6 show how much improvement has been made compared to the DDPG baseline, and how each of the three proposed schemes affects the results.

TABLE II
THE NUMBER OF SIMULATIONS WHEN FoM CONVERGES TO ZERO

Methods	Basic two-stage amplifier	Telescopic amplifier	Folded cascode amplifier
DDPG+UVFA	10,110	14,059	7,743
DDPG+UVFA +GI	8,836	9,119	5,605
DDPG+UVFA +GI+BS	**5,843**	**8,508**	**4,840**

*BO [3] and DDPG [13] did not converge within 10 hours.

When we compare DDPG and DDPG+UVFA, UVFA tends to find well by adding a vector that act as flags to the DDPG baseline, and it served as a guideline. Furthermore, it is simpler because the modified UVFA we proposed can reduce dimension that applying UVFA as it is. Next, it can be seen that the circuit optimized with GI, which was using g_m/I_D factors that changed the framework, is relatively better at convergence and also converges within a short time. The agent tended to learn faster because the non-linearity for circuit performance was reduced compared to the existing method with optimizing with geometrical design variable. Finally, the BS for PZ sampling and pick sampling, including random sampling, enables the network to train more efficiently. This helps the approximator learn by distinguishing between state and goal better in a situation where state and goal are concatenated.

Fig. 6 is a graph of how fast the max value of the FoM was reached to zero as the simulation progressed. In the case of BO or DDPG baselines trained based on the existing geometrical design variable, the target circuit performance was not found, but it is shown that it can be improved through UVFA. Furthermore, it can be seen that learning with GI, that is, electrical design variable-based data, which is g_m/I_D, is more effective than learning with geometrical design variable-based data. Finally, DDPG+UVFA+GI+BS, which has applied all we proposed, improved by 42.2%, 39.5%, and 37.5%, respectively, for the three circuits compared to DDPG+UVFA.

V. CONCLUSION

This study proposed a novel framework for analog circuit optimization with efficient search algorithm: 1) sensitivity analysis between g_m/I_D and circuit performance, which reduces non-linearity, was applied to achieve better training ability; 2) UVFA was applied more simply by designating the goal as the FoM equation, which is a vector that represents the state; and 3) the data were selected and sampled according to the criteria set with circuit design knowledge. The optimal design was not identified for BO or DDPG baseline algorithm. However, it's been strengthened by modified UVFA, resulting in convergence. Using the electrical design variable-based data, g_m/I_D, based on circuit performance sensitivity analysis matrix, and novel buffer sampling method, we improved up to 42.2%. By using proposed methods, designers do not have to perform sizing manually, and the solution can be obtained automatically in a few hours.

ACKNOWLEDGMENT

The EDA Tool was supported by the IC Design Education Center. This work was supported by Institute of Information communications Technology Planning Evaluation (IITP) grant funded by the Korea government(MSIT) (No.2021-0-00754, Software Systems for AI Semiconductor Design). This work was supported by Institute of Information communications Technology Planning Evaluation (IITP) grant funded by the Korea government (MSIT).(No.2022-0-01172, DRAM PIM Design Base Technology Development). This work was supported by Samsung Electronics Co., Ltd.

REFERENCES

[1] Zhigang Ji, Haibao Chen, and Xiuyan Li. Design for reliability with the advanced integrated circuit (ic) technology: challenges and opportunities. *Science China Information Sciences*, 62(12):1–4, 2019.

[2] Fabio D'Agostino and Daniele Quercia. Short-channel effects in mosfets. *Introduction to VLSI design (EECS 467)*, 70:71–72, 2000.

[3] Wenlong Lyu, Pan Xue, Fan Yang, Changhao Yan, Zhiliang Hong, Xuan Zeng, and Dian Zhou. An efficient bayesian optimization approach for automated optimization of analog circuits. *IEEE Transactions on Circuits and Systems I: Regular Papers*, 65(6):1954–1967, 2018.

[4] Shuhan Zhang, Fan Yang, Dian Zhou, and Xuan Zeng. An efficient asynchronous batch bayesian optimization approach for analog circuit synthesis. In *2020 57th ACM/IEEE Design Automation Conference (DAC)*, pages 1–6, 2020.

[5] Zhenxin Zhao and Lihong Zhang. Deep reinforcement learning for analog circuit sizing. In *2020 IEEE International Symposium on Circuits and Systems (ISCAS)*, pages 1–5. IEEE, 2020.

[6] NS Karthik Somayaji, Hanbin Hu, and Peng Li. Prioritized reinforcement learning for analog circuit optimization with design knowledge. In *2021 58th ACM/IEEE Design Automation Conference (DAC)*, pages 1231–1236. IEEE, 2021.

[7] Keertana Settaluri, Ameer Haj-Ali, Qijing Huang, Kourosh Hakhamaneshi, and Borivoje Nikolic. Autockt: Deep reinforcement learning of analog circuit designs. In *2020 Design, Automation & Test in Europe Conference Exhibition (DATE)*, pages 490–495, 2020.

[8] Hanrui Wang, Kuan Wang, Jiacheng Yang, Linxiao Shen, Nan Sun, Hae-Seung Lee, and Song Han. Gcn-rl circuit designer: Transferable transistor sizing with graph neural networks and reinforcement learning. In *2020 57th ACM/IEEE Design Automation Conference (DAC)*, pages 1–6. IEEE, 2020.

[9] Bobak Shahriari, Kevin Swersky, Ziyu Wang, Ryan P. Adams, and Nando de Freitas. Taking the human out of the loop: A review of bayesian optimization. *Proceedings of the IEEE*, 104(1):148–175, 2016.

[10] Paul Jespers. *The gm/ID Methodology, a sizing tool for low-voltage analog CMOS Circuits: The semi-empirical and compact model approaches*. Springer Science & Business Media, 2009.

[11] F. Silveira, D. Flandre, and P.G.A. Jespers. A gm/id based methodology for the design of cmos analog circuits and its application to the synthesis of a silicon-on-insulator micropower ota. *IEEE Journal of Solid-State Circuits*, 31(9):1314–1319, 1996.

[12] H Daoud Dammak, S Bensalem, S Zouari, and M Loulou. Design of folded cascode ota in different regions of operation through gm/id methodology. *International Journal of Electrical and Electronics Engineering*, 1(3):178–183, 2008.

[13] Timothy P Lillicrap, Jonathan J Hunt, Alexander Pritzel, Nicolas Heess, Tom Erez, Yuval Tassa, David Silver, and Daan Wierstra. Continuous control with deep reinforcement learning. *arXiv preprint arXiv:1509.02971*, 2015.

[14] Tom Schaul, Daniel Horgan, Karol Gregor, and David Silver. Universal value function approximators. In Francis Bach and David Blei, editors, *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 1312–1320, Lille, France, 07–09 Jul 2015. PMLR.

[15] S Devi, Gourav Tilwankar, and Rajesh Zele. Automated design of analog circuits using machine learning techniques. In *2021 25th International Symposium on VLSI Design and Test (VDAT)*, pages 1–6, 2021.

[16] Yannick Uhlmann, Michael Essich, Lennart Bramlage, Jürgen Scheible, and Cristóbal Curio. Deep reinforcement learning for analog circuit sizing with an electrical design space and sparse rewards. In *2022 ACM/IEEE 4th Workshop on Machine Learning for CAD (MLCAD)*, pages 21–26, 2022.

[17] Marcin Andrychowicz, Filip Wolski, Alex Ray, Jonas Schneider, Rachel Fong, Peter Welinder, Bob McGrew, Josh Tobin, OpenAI Pieter Abbeel, and Wojciech Zaremba. Hindsight experience replay. *Advances in neural information processing systems*, 30, 2017.

[18] Zihong Liu, Chao Bian, Zhihua Wang, and Chun Zhang. Full custom design of a two-stage fully differential cmos amplifier with high unity-gain bandwidth and large dynamic range at output. In *48th Midwest Symposium on Circuits and Systems, 2005.*, pages 984–987. IEEE, 2005.

[19] Mo M Zhang and Paul J Hurst. Effect of nonlinearity in the cmfb circuit that uses the differential-difference amplifier. In *2006 IEEE International Symposium on Circuits and Systems*, pages 4–pp. IEEE, 2006.

[20] RATNAPRABHA W Jasutkar, PR Bajaj, and AY Deshmukh. Design of 1v, 0.18 μ folded cascode operational amplifier for switch capacitor sigma delta modulator. *Int. J. Elect. Electron. Eng. Res*, 3:223–232, 2013.